

Calculus 2 - Homework 3 (Problems)

1.

Determine whether $\int_0^{\infty} x e^{-x} dx$ converges, and if so find its value.

2.

Classify $\sum_{n=1}^{\infty} (-1)^n \cdot n/(n+1)$ as absolutely convergent, conditionally convergent, or divergent.

3.

Classify $\sum_{n=1}^{\infty} (-1)^n/\sqrt{n}$ as absolutely convergent, conditionally convergent, or divergent.

4.

Evaluate $\int \sin^2(x) dx$.

5.

Use Simpson's Rule with $n = 4$ to approximate $\int_0^1 1/(1+x^2) dx$. Report S_4 to 6 decimals. (The exact value is $\arctan 1 = \pi/4$.)

6.

Determine whether $\sum_{n=1}^{\infty} (n+2)/(n^3+n)$ converges or diverges.

7.

For the convergent alternating series $\sum_{n=1}^{\infty} (-1)^{n+1}/n^3$, use the Alternating Series Remainder bound $|R_N| \leq b_{N+1}$ to find the smallest number of terms N needed so the partial sum approximates the total with error less than 0.01.

8.

Use the Trapezoidal Rule with $n = 4$ to approximate $\int_1^2 (1/x) dx$. Round node values to 4 decimals and report T_4 to 5 decimals.

9.

Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} n!/n^n$ converges or diverges.

10.

Evaluate $\int x^2/\sqrt{4-x^2} dx$.